

Determining the Fairness of a Dice with the Help of Chi-Square Test

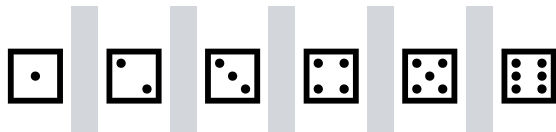
ENG121 Mechanics Lab Section D

12th February, 2025

1. Introduction

The experiment is all about checking if a regular six-sided dice is fair or not, and whether some numbers show up more frequently than others. It is known that a die has six faces, respectfully (“1”, “2”, “3”, “4”, “5”, “6”), representing six possible outcomes. The null hypothesis is that the probability of each face appearing is assumed to be $1/6$, because the die after being thrown can show only on of the six possible outcomes, so they are assumed to be equally likely to occur. On the other hand, the alternative hypothesis is that the holes on the surface of the dice unbalance the center of mass, making the probabilities unequal. Chi Square Test will be used to test the hypothesis.

Sample Space:



(Figure 1.1 Sample Space)

2. Measurements

Procedure

Step 1) A group of 10 dice tosses per roll.

Step 2) Face frequencies measured and recorded accordingly in *(See Figure 1.2)*

Step 3) Steps 1) and 2) repeated 21 times.

Step 4) Frequencies for each face computed for the first 6 measurements, 12 measurements and 21 measurements. *(See Figure 1.3)*

Step 5) Probability distributions of the gathered data for the first 6 measurements, 12 measurements and 21 measurements on the same axis. (See Figure 1.4)

Step 6) Chi Square Test applied to compare with expected values.

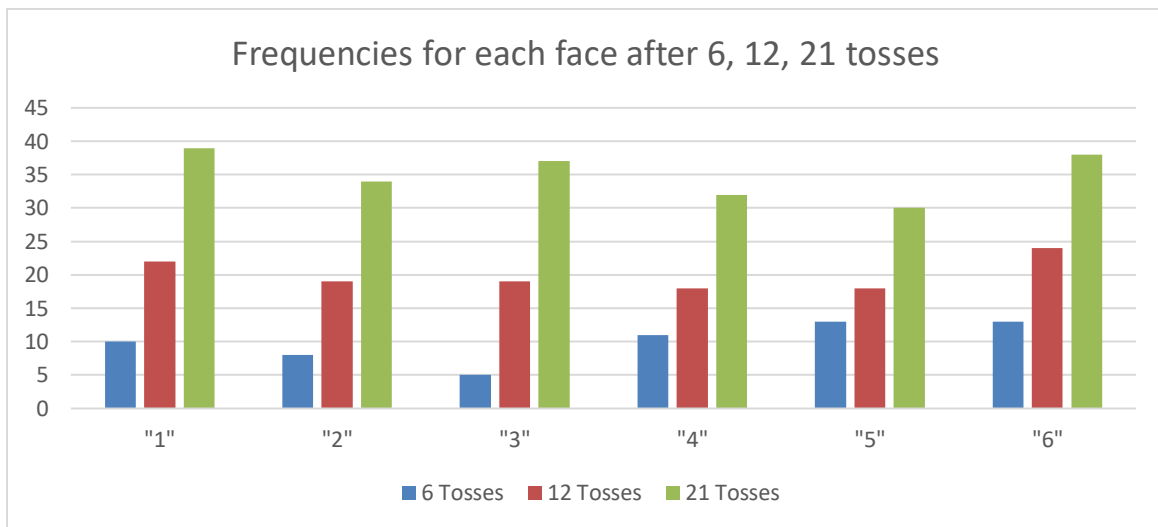
Frequency	"1"	"2"	"3"	"4"	"5"	"6"
1 st toss	2	2	0	4	0	2
2 nd toss	1	2	1	3	2	1
3rd toss	2	1	0	0	5	2
4th toss	2	2	1	1	1	3
5th toss	1	1	2	3	1	2
6th toss	2	0	1	0	4	3
7th toss	3	3	1	1	1	1
8th toss	0	1	6	1	0	2
9th toss	2	2	2	1	0	3
10nd toss	4	2	1	0	2	1
11nd toss	2	3	1	2	0	2
12nd toss	1	0	3	2	2	2
13nd toss	0	3	3	2	1	1
14th toss	0	0	4	1	2	3
15th toss	1	3	2	2	1	1
16th toss	2	1	2	3	2	0
17th toss	2	1	1	2	2	2
18th toss	4	2	1	1	1	1
19th toss	0	3	2	1	1	3
20nd toss	3	1	2	1	1	2
21st toss	5	1	1	1	1	1
Total frequency	39	34	37	32	30	38
Observed probability	0.186	0.162	0.176	0.152	0.143	0.181

(Figure 1.2 Recorded frequencies of each possible face)

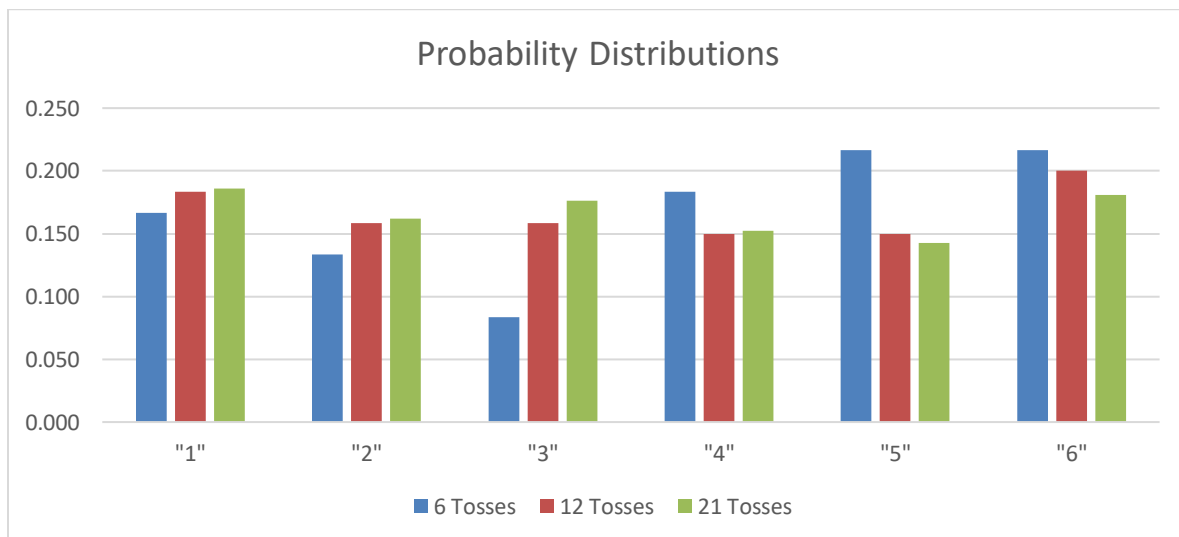
Errors

Name	Type	Impact	Solution
Dice Imperfections	Systematic	Slight imbalance in dice's mass distribution can lead to some faces showing up more often than other face.	New dice to be used, but no factory

Tossing Height	Systematic	Dice tossed from a higher distance makes the dice outcome more randomized compared to the one dropped from a lower level.	Optimal height for tossing the dice to be selected ensuring random results.
Human Error in Counting Occurrences	Blunder	When in a hurry there are chances to miscount the dice outcomes.	Measurements double checked before recording.
Surface Type	Systematic	When the dice touches the surface, the friction makes the dice bounce chaotically.	Optimal surface selected that meets the requirements.



(Figure 1.3 Recorded frequencies of each face after 6, 12, 21 tosses)



(Figure 1.4 Probability distributions for first 6, 12, 21 tosses)

Instruments

Name	Quantity	Properties
Die	10	Plastic, ~8 mm * 8 mm * 8 mm

3. Data Analysis

Expected Probability for Each Face

$$P = 1/6$$

Degree of Freedom

The Degree of Freedom can be calculated by knowing that the result of five cases can be used to calculate the remaining sixth one, hence **$df = \text{Number of categories} - 1 = 6 - 1 = 5$** .

Significance Level

$P_{df}(\chi^2) = 0.05$ meaning there is a 5% chance of rejecting the null hypothesis, from the Chi Square Distribution Table the critical data for one tailed test at **$df = 5$** and **$P_{df}(\chi^2) = 0.05$** can be found.

$$\chi^2_{0.05} = 11.070$$

Chi Square Test

$$\chi^2 = \sum_{i=1}^6 \frac{(O_i - E_i)^2}{E_i}$$

After applying the Chi Square Formula (See Formula 1.1) on values for 6, 12, 21 tosses the χ^2 values are calculated to be accordingly 4.8, 1.5, 1.828.

	Observed Number. (O)	Expected Number (E)	O – E	$(O - E)^2$	$\frac{(O - E)^2}{E}$
“1”	10	10	0	0	0
“2”	8	10	-2	4	0.4
“3”	5	10	-5	25	2.5
“4”	11	10	1	1	0.1
“5”	13	10	3	9	0.9
“6”	13	10	3	9	0.9
<i>Total</i>	60	60			4.8

(Figure 1.5 Chi Square for first 60 dice rolls.)

	Observed Number. (O)	Expected Number (E)	O – E	$(O - E)^2$	$\frac{(O - E)^2}{E}$
“1”	22	20	2	4	0.2
“2”	19	20	-1	1	0.05
“3”	19	20	-1	1	0.05
“4”	18	20	-2	4	0.2
“5”	18	20	-2	4	0.2
“6”	24	20	4	16	0.8
<i>Total</i>	120	120			1.5

(Figure 1.5 Chi Square for first 120 dice rolls.)

	Observed Number. (O)	Expected Number (E)	O – E	$(O - E)^2$	$\frac{(O - E)^2}{E}$
“1”	39	35	4	16	0.457
“2”	34	35	-1	1	0.029
“3”	37	35	2	4	0.114
“4”	32	35	-3	9	0.257
“5”	30	35	-5	25	0.714
“6”	38	35	3	9	0.257
<i>Total</i>	210	210			1.828

(Figure 1.5 Chi Square for the first 210 rolls.)

4. Conclusion and Evaluation

The value $\chi^2 = 1.828$ is lower than the critical value of $\chi_{0.05}^2 = 11.070$, there is not enough evidence to support the alternative hypothesis, so the null hypothesis is reasonable at the 5% significance level.

In conclusion, the experiment was held to test whether the center of mass imbalance caused by the holes on die's surface makes it unfair. After measuring, recording, analyzing the observed data, it is concluded that the probabilities for each face to occur are equivalent and equal to 1/6 and the holes on die's faces do not have a significant impact on it's performance.

5. References

Chi-Square distribution Table. (n.d.). In *Chi-Square Distribution Table* (p. 2).

<https://www.math.arizona.edu/~jwatkins/chi-square-table.pdf>